components work together with the software system to ensure accurate coin detection, safe electrical operation, and reliable execution of printing tasks.

Related Literature and Studies

Self-Service Kiosk Technologies

According to Arevalo et al. (2025), Efficient Printing Solutions through the Automated Kiosk System, the demand for accessible and efficient printing services in academic institutions has increased due to the large number of students requiring printed academic materials such as assignments, reports, and research papers. The study investigated the development of an automated printing kiosk system designed to address the issue of long queues and delays commonly experienced in campus printing shops. Based on their survey conducted at Adamson University, students were reported to wait between one and twenty minutes just to print short documents, highlighting the inefficiency of traditional printing services. To address this issue, the researchers proposed a QR code–based automated printing kiosk that allows users to upload documents, configure print settings, and generate a QR code ticket through a web application, which can then be scanned at the kiosk to initiate printing. The findings of the study emphasized that automated kiosk systems significantly improve service efficiency by minimizing manual intervention and streamlining the document submission process. These results are directly relevant to the development of the PrintBit: A Tablet-Based Self-Service Printing Kiosk Machine, as both systems aim to provide a more efficient and accessible printing solution for students. Similar to the proposed automated kiosk in the study, the PrintBit system enables users to upload files wirelessly and process printing transactions independently through a kiosk interface, thereby improving the availability and convenience of document printing services within Laguna University.

According to Anduyo (2025), self-service kiosk systems enable users to perform transactions independently through automated interfaces without requiring assistance from service staff. the adoption of automated kiosk technologies significantly improves operational efficiency and service accessibility in environments where large numbers of users need to access services quickly. The study examined factors influencing the use of automated kiosk systems and found that kiosks reduce waiting time by allowing users to complete transactions independently without relying on staff assistance. In addition, kiosk systems